

BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, PILANI
INSTRUCTION DIVISION
First Semester 2026-2027
Course Handout (Part II)

In addition to part-I (General handout for all courses appended to the timetable) this portion gives further specific details regarding the course:

1. a) Course Number: CS F446

b) Course Title: Data Storage Technologies and Networks

c) Instructors: **J.P. Mishra (IC)** (email: **jpm**)

2. a) Objective: To introduce the student to secondary storage technologies, algorithms and protocols for effective access of secondary storage, design and implementation issues for high performance storage over the network, and performance analysis as applicable.

b) Scope: The course covers storage systems ranging from RAID to NAS to SAN as well as their architecture and implementation. File systems and application scenarios are also covered. Rudimentary performance studies are also included.

3. Text and Reference Books:

(a) Text Book:

T1. Storage Networks: The complete Reference. Robert Spalding TMH.

(b) References:

R1. Computer Systems – A Programmer’s Perspective. Randall Bryant and David O’Hallaron. Pearson Education. 2003.

R2. The Design and Implementation of the 4.4 BSD Operating System. McKusick, Bostic, Karels, and Quateman. 1996

R3. Computer Architecture. Patterson and Hennessey. 3rd Edition. 2007

R4. IBM Redbooks on SAN, SAN Design and Storage Virtualization. (Available online)

AR. Additional references as assigned and made available by the instructors (Papers, Standards Documents and Datasheets).

4. Course Plan

Lecture No.s	Topic	Reading / Reference
1	Role of Data in Computing; Storage Requirements	T1. Ch. 1
2	Storage Models. Large Data Cases; Data Explosion; Data and Storage Characteristics.	AR
3-11	Network basics : Collision domain, Broadcast domain , network segmentation using Hub, bridge, Routers,, Switching modes, High availability network : STP	Class notes
12	Computer System Architecture – I/O: Buses, Design Issues and Typical Configurations.	AR
13 – 14	Hard Disks – Structure and Geometry, Addressing and Access Model. Access Time	R1. Ch 6
15	SCSI Protocol – Bus Features and State Transitions. Commands.	AR

16 – 17	Disk I/O Scheduling: Advanced Techniques. Performance Issues, Prefetching/Caching.	AR
18 – 19	Storage Design – Reliability and Redundancy. Performance: Data Transfer Rate vs. I/O Rate. Striping and Mirroring. Error Detection and Correction	AR
20 – 21	RAID –Levels and Performance. Parity Placement.	R3, AR
22	RAID Controllers – Architecture, Caching and Destaging.	AR
23 – 24	Fibre Channel – Layered Architecture, Topologies, Classes of Service.	R4 T1. Ch 11
25 – 26	Fibre Channel – Addressing and Routing. High Performance Techniques.	R4 T1. Ch 11
27	Storage on the Network. Evolution. SAN vs. NAS	T1. Ch. 14
28 – 29	SAN Structure and Components. SAN Performance Issues and Techniques.	R4
30	SAN Switches – Architecture and Case Study	AR
31 – 32	SAN Extensions and Emerging Protocols – IP SAN. SAN Solutions – Storage Virtualization	R4, AR
33	NAS – Structure and Components, NAS Servers and Configurations	T1. Ch 10
34 – 35	NAS Filesystems	R3 Ch 9, AR
36 – 37	Advanced FileSystems – Techniques and Case Studies	AR
38 – 40	Storage Architectures and Solutions. Data Centers – Issues, Techniques and Performance.	T1. Ch 14. AR

4. Evaluation Scheme:

Component	Type	Duration	Date&Time	Weight
Mid Sem	Closed Book	90 mins	07/10 FN1	35%
Term paper	Take Home	2 to 3 weeks each	-	25%
Comprehensive Exam	Part Open Book	3 Hrs	08/12 FN	40%

5. Notices:

All notices will be Posted on Naland.Watch out for class announcements.

6. Chamber Consultation Hours: W- 5 to 6 PM

Instructor-in-charge